THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE CHAPTER TEST SET A

Chapter 10: **Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An object that produces its own light is called: 1

(a) luminous **(b)** opaque **(c)** non-luminous **(d)** transparent

2. Materials that allow light to pass completely are: 1

(a) transparent **(b)** opaque **(c)** luminous **(d)** translucent

3. Which is NOT needed for a shadow? 1

(a) a mirror **(b)** light source **(c)** opaque object **(d)** screen

4. The bouncing back of light from a surface is: 1

(a) absorption **(b)** reflection **(c)** dispersion **(d)** refraction

5. The left-right reversal in a mirror is called: 1

(a) rotation **(b)** refraction
(c) lateral inversion **(d)** dispersion

6. Image in a pinhole camera is ___ if the hole is large. 1

(a) sharp **(b)** smaller **(c)** brighter only **(d)** blurred

7. In a kaleidoscope, mirrors are placed at: 1

(a) flat on table **(b)** perpendicular to the Sun
(c) an angle to each other **(d)** far apart parallel

8. A rough surface causes: 1

(a) regular reflection **(b)** no light
(c) diffused reflection **(d)** clear images

9. A mirror image of 'b' appears as: 1

(a) p upside **(b)** q
(c) same b **(d)** d-like (laterally inverted)

10. The pinhole camera works because light: 1

(a) is absorbed **(b)** travels in straight lines
(c) stops at hole **(d)** bends strongly

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 15–25 words each.

11. Differentiate luminous and non-luminous objects with examples. 2

12. Distinguish transparent, translucent and opaque materials. 2

13. What three things are needed to form a shadow? 2

14. Why does a shadow of an opaque object have the shape of the object? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe an activity to show that light travels in a straight line. 3

16. How does the size of a shadow change with the distance of the object from the source? 3

17. Explain how a shadow is formed. 3

OR

Why do translucent objects make lighter shadows than opaque ones? 3

18. State and explain the characteristics of an image in a plane mirror. 3

OR

Describe how a pinhole camera is made. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) Define luminous and non-luminous bodies.
(ii) Give two examples of each.
(iii) Why is the Moon visible? 5

OR

Case study: Asha places a ball between a torch and a wall and observes a shadow. 5
(i) Why does it form?
(ii) What if the ball moves closer to the torch?
(iii) What if it is replaced by tracing paper?
(iv) Which property of light is shown?

20. (i) Classify materials by transmission of light with examples.
(ii) Which makes dark shadows?
(iii) Why? 5

OR

Case study: A girl holds a card with 'ROSE' in front of a mirror. 5
(i) How does it appear?
(ii) Name the effect.
(iii) Distance of image if card is 40 cm away.
(iv) Nature of image.

END OF QUESTION PAPER — Total: 40 Marks

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